Solutions of Homework 02

Answer 1

IDENTIFY: Given two displacements, we want the average velocity and the average speed.

SET UP: The average velocity is $v_{\text{av-}x} = \dfrac{\Delta x}{\Delta t}$ and the average speed is just the total distance walked divided by the total time to walk this distance.

EXECUTE: **(a)** Let $+x$ be east. $\Delta x = 60.0\text{ m} - 40.0\text{ m} = 20.0\text{ m}$ and $\Delta t = 28.0\text{ s} + 36.0\text{ s} = 64.0\text{ s}$. So $v_{\text{av-}x} = \dfrac{\Delta x}{\Delta t} = \dfrac{20.0\text{ m}}{64.0\text{ s}} = 0.312\text{ m/s}$.

(b) average speed $= \dfrac{60.0\text{ m} + 40.0\text{ m}}{64.0\text{ s}} = 1.56\text{ m/s}$

EVALUATE: The average speed is much greater than the average velocity because the total distance walked is much greater than the magnitude of the displacement vector.

Answer 2

a)

$$\vec{r}(t) = x(t)\hat{\imath} + y(t)\hat{\jmath} = \alpha t\hat{\imath} + (3.0 - \beta t^2)\hat{\jmath}$$

The velocity vector is:

$$\vec{v}(t) = \frac{dx(t)}{dt}\hat{\imath} + \frac{dy(t)}{dt}\hat{\jmath}$$

$$\vec{v}(t) = \alpha\hat{\imath} - 2\beta t\hat{\jmath}$$

The acceleration vector is:

$$\vec{a}(t) = -2\beta\hat{\jmath}$$

(b) The magnitude of the velocity vector is, at t=2.0 s:

$$v(t = 2.0\ s) = |\vec{v}(t = 2.0\ s)| = \sqrt{{v_x}^2 + {v_y}^2}$$

$$v(t = 2.0\ s) = \sqrt{\alpha^2 + 16\beta^2} = \sqrt{2.4^{\ 2} + 16 \times 1.2^{\ 2}}$$

$$v(t = 2.0\ s) = 5.37\ m/s$$

The magnitude of the acceleration vector is:

$$a(t) = |\vec{a}(t)| = 2\beta = 2.4\ m/s$$

Comment.

For a motion in the xy-plane:

$$a = \sqrt{{a_x}^2 + {a_y}^2}$$

But the acceleration vector is here directed toward the $-y$-direction, thus $a_x = 0$ and

$$a = \sqrt{{a_y}^2} = |a_y|$$

Answer 3

IDENTIFY and **SET UP:** The instantaneous velocity is the slope of the tangent to the x versus t graph.
EXECUTE: **(a)** The velocity is zero where the graph is horizontal; point IV.
(b) The velocity is constant and positive where the graph is a straight line with positive slope; point I.
(c) The velocity is constant and negative where the graph is a straight line with negative slope; point V.
(d) The slope is positive and increasing at point II.
(e) The slope is positive and decreasing at point III.
EVALUATE: The sign of the velocity indicates its direction.

Answer 4

SET UP: Assume the ball starts from rest and moves in the $+x$-direction.
EXECUTE: **(a)** $x - x_0 = 1.50$ m, $v_x = 45.0$ m/s and $v_{0x} = 0$. $v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$ gives

$$a_x = \frac{v_x^2 - v_{0x}^2}{2(x - x_0)} = \frac{(45.0\text{ m/s})^2}{2(1.50\text{ m})} = 675\text{ m/s}^2.$$

(b) $x - x_0 = \left(\frac{v_{0x} + v_x}{2}\right)t$ gives $t = \frac{2(x - x_0)}{v_{0x} + v_x} = \frac{2(1.50\text{ m})}{45.0\text{ m/s}} = 0.0667$ s

EVALUATE: We could also use $v_x = v_{0x} + a_x t$ to find $t = \frac{v_x}{a_x} = \frac{45.0\text{ m/s}}{675\text{ m/s}^2} = 0.0667$ s which agrees with

Answer 5

SET UP: Assume the ball moves in the $+x$ direction.
EXECUTE: **(a)** $v_x = 73.14$ m/s, $v_{0x} = 0$ and $t = 30.0$ ms. $v_x = v_{0x} + a_x t$ gives

$$a_x = \frac{v_x - v_{0x}}{t} = \frac{73.14\text{ m/s} - 0}{30.0 \times 10^{-3}\text{ s}} = 2440\text{ m/s}^2.$$

(b) $x - x_0 = \left(\frac{v_{0x} + v_x}{2}\right)t = \left(\frac{0 + 73.14\text{ m/s}}{2}\right)(30.0 \times 10^{-3}\text{ s}) = 1.10$ m

EVALUATE: We could also use $x - x_0 = v_{0x}t + \frac{1}{2}a_x t^2$ to calculate $x - x_0$:

$x - x_0 = \frac{1}{2}(2440\text{ m/s}^2)(30.0 \times 10^{-3}\text{ s})^2 = 1.10$ m, which agrees with our previous result. The acceleration of the ball is very large.

Answer 6

IDENTIFY: The car moves in projectile motion. The car travels $21.3\text{ m} - 1.80\text{ m} = 19.5\text{ m}$ downward during the time it travels 61.0 m horizontally.

SET UP: Take $+y$ to be downward. $a_x = 0$, $a_y = +9.80\text{ m/s}^2$. $v_{0x} = v_0$, $v_{0y} = 0$.

EXECUTE: **(a)** Use the vertical motion to find the time in the air:

$$y - y_0 = v_{0y}t + \tfrac{1}{2}a_y t^2 \text{ gives } t = \sqrt{\frac{2(y - y_0)}{a_y}} = \sqrt{\frac{2(19.5\text{ m})}{9.80\text{ m/s}^2}} = 1.995\text{ s}$$

Then $x - x_0 = v_{0x}t + \tfrac{1}{2}a_x t^2$ gives $v_0 = v_{0x} = \dfrac{x - x_0}{t} = \dfrac{61.0\text{ m}}{1.995\text{ s}} = 30.6\text{ m/s}.$

(b) $v_x = 30.6\text{ m/s}$ since $a_x = 0$. $v_y = v_{0y} + a_y t = -19.6\text{m/s}$. $v = \sqrt{v_x^2 + v_y^2} = 36.3\text{m/s}.$

EVALUATE: We calculate the final velocity by calculating its x and y components.

Answer 7

IDENTIFY: Uniform circular motion.

SET UP: Since the magnitude of $\vec{v}$ is constant, $v_{\tan} = \dfrac{d|\vec{v}|}{dt} = 0$ and the resultant acceleration is equal to the radial component. At each point in the motion the radial component of the acceleration is directed in toward the center of the circular path and its magnitude is given by v^2/R.

EXECUTE: **(a)** $a_{\text{rad}} = \dfrac{v^2}{R} = \dfrac{(7.00\text{ m/s})^2}{14.0\text{ m}} = 3.50\text{ m/s}^2$, upward.

(b) The radial acceleration has the same magnitude as in part (a), but now the direction toward the center of the circle is downward. The acceleration at this point in the motion is 3.50 m/s^2, downward.

(c) SET UP: The time to make one rotation is the period T, and the speed v is the distance for one revolution divided by T.

EXECUTE: $v = \dfrac{2\pi R}{T}$ so $T = \dfrac{2\pi R}{v} = \dfrac{2\pi\,(14.0\text{ m})}{7.00\text{ m/s}} = 12.6\text{ s}$

EVALUATE: The radial acceleration is constant in magnitude since v is constant and is at every point in the motion directed toward the center of the circular path. The acceleration is perpendicular to $\vec{v}$ and is nonzero because the direction of $\vec{v}$ changes.

Answer 8

SET UP: $T = 24$ h.

EXECUTE: **(a)** $a_{\text{rad}} = \dfrac{4\pi^2(6.38\times10^6\ \text{m})}{((24\ \text{h})(3600\ \text{s/h}))^2} = 0.034\ \text{m/s}^2 = 3.4\times10^{-3}\,g.$

(b) Solving Eq. (3.30) for the period T with $a_{\text{rad}} = g$, $T = \sqrt{\dfrac{4\pi^2(6.38\times10^6\,\text{m})}{9.80\ \text{m/s}^2}} = 5070\ \text{s} = 1.4\ \text{h}.$

EVALUATE: a_{rad} is proportional to $1/T^2$, so to increase a_{rad} by a factor of $\dfrac{1}{3.4\times10^{-3}} = 294$ requires that T be multiplied by a factor of $\dfrac{1}{\sqrt{294}}$. $\dfrac{24\ \text{h}}{\sqrt{294}} = 1.4\ \text{h}.$

Answer 9

IDENTIFY: Apply the relative velocity relation.

SET UP: The relative velocities are $\vec{v}_{\text{C/E}}$, the canoe relative to the earth, $\vec{v}_{\text{R/E}}$, the velocity of the river relative to the earth and $\vec{v}_{\text{C/R}}$, the velocity of the canoe relative to the river.

EXECUTE: $\vec{v}_{\text{C/E}} = \vec{v}_{\text{C/R}} + \vec{v}_{\text{R/E}}$ and therefore $\vec{v}_{\text{C/R}} = \vec{v}_{\text{C/E}} - \vec{v}_{\text{R/E}}$. The velocity components of $\vec{v}_{\text{C/R}}$ are $-0.50\ \text{m/s} + (0.40\ \text{m/s})/\sqrt{2}$, east and $(0.40\ \text{m/s})/\sqrt{2}$, south, for a velocity relative to the river of 0.36 m/s, at $52.5°$ south of west.

EVALUATE: The velocity of the canoe relative to the river has a smaller magnitude than the velocity of the canoe relative to the earth.